

Wannes Fraj

AI Developer & Robotics Engineer

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SUMMARY

Industrial Computer Engineering student at ENET'Com with a dual expertise in Artificial Intelligence and Robotics. Experienced in building end-to-end AI systems — from ML pipelines and LLM orchestration to embedded firmware and autonomous navigation. Passionate about intelligent systems at the edge, combining AI-driven perception with real-time robotic control. Seeking a summer internship in AI, robotics, or intelligent automation engineering.

EDUCATION

- National Engineering School of Sfax (ENET'Com)** 2025 – Present
Industrial Computer Engineering Sfax, Tunisia
- Faculty of Sciences of Monastir** 2023 – 2025
Preparatory Cycle in Physics and Chemistry Monastir, Tunisia

TECHNICAL STACK

- Languages:** Python, C, C++, Java, MATLAB, Dart, JavaScript
- AI & ML:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Deep Learning, Computer Vision, NLP
- LLM & Agentic Systems:** LangChain, LlamaIndex, RAG, Ollama, LLM Orchestration, N8N
- Robotics & Navigation:** ROS2, Nav2, SLAM, LiDAR, RViz, Gazebo, Path Planning, Sensor Fusion
- Embedded Systems & IoT:** ESP32, MQTT, Mosquitto, BLE, PlatformIO, Motor Control, Real-Time Firmware
- Data & Simulation:** NumPy, Pandas, Matplotlib, Jupyter, Proteus 8, Feature Engineering

PROJECTS

- EventPlanner – Agentic AI Chatbot** (2nd Place – N8N Hackathon) 2026
Tools: N8N, LLMs, Agentic Workflows, REST APIs [\[G\]](#)
 - Built an agentic event-planning chatbot automating end-to-end event creation via multi-step LLM orchestration in N8N, handling intent, slot filling, and real-time API retrieval.
 - Awarded **2nd place** in a competitive hackathon focused on AI workflow automation.
- Autonomous Heterogeneous Vehicle Navigation** (3rd Place – IEEE TSYP 2025) 2025
Tools: ROS2, Nav2, LiDAR, SLAM, Gazebo, RViz [\[in\]](#)
 - Developed an autonomous navigation system using ROS2 Nav2 with SLAM and LiDAR for real-time mapping, localization, and obstacle avoidance in dynamic environments.
 - Awarded **3rd place** at the IEEE TSYP 2025 national robotics competition.
- CubeSat Thermal Prediction – Embedded AI for Space Systems** 2025
Tools: Python, LSTM, Operational Research, Synthetic Dataset [\[G\]](#)
 - Trained an LSTM to predict solar-facing panel orientations for thermal risk management using synthetic orbital datasets simulating real-world attitude changes.
 - Applied operational research algorithms for orientation classification, improving mission reliability and autonomous decision-making.
- Smart Home – Wireless Roller Shutter Control System** 2026
Tools: ESP32, C++, MQTT, Mosquitto, BLE, Flutter, Dart, PlatformIO, Proteus 8 [\[G\]](#)
 - Developed ESP32 firmware (BLE provisioning, MQTT control, manual override, percentage-based positioning) and a cross-platform Flutter app with real-time control.
 - Designed full circuit schematic and simulated in Proteus 8.

CERTIFICATIONS

- Deep Learning – NVIDIA DLI** 2026

VOLUNTEERING

- Webmaster** Jan 2026 – Present
IEEE ENET'Com WIE Student Affinity Group
 - Designed visual content (posters, banners) using Photoshop and Premiere Pro.
 - Coordinated event logistics and team communication for workshops and tech events.
 - Managed online presence and publication schedules across platforms.